Random Variables and Probability Distributions

Random Variables

- Question: A coin is tossed three times; what is the probability you will get two "heads"?
- A random variable (RV) is "something that takes on numerical values depending on chance"
- Usually denoted by the capital letters X, Y, Z
- Examples
  - Value of a stock next month
  - Time I'll wait for the subway today
  - Number of iPods to be sold this year

Tossing a Coin Three Times

- Define: X = number of "heads"

Outcome	Prob.	X
H H H	1/8	3
H H T	1/8	2
H T H	1/8	2
H T T	1/8	1
T H H	1/8	2
T H T	1/8	1
T T H	1/8	1
T T T	1/8	0

$\Rightarrow$

X	P(X = x)
0	1/8
1	3/8
2	3/8
3	1/8

Discrete vs. Continuous RVs

- Discrete RV: No “in between” values; often “counting”
  - Number of “heads”
  - Number of iPods sold
  - Sum of two dice rolls
- Continuous RV: Always “in between” values; often “measuring”
  - Waiting time for subway
  - Tomorrow’s temperature
  - Value of a stock next month

The Probability Distribution Function

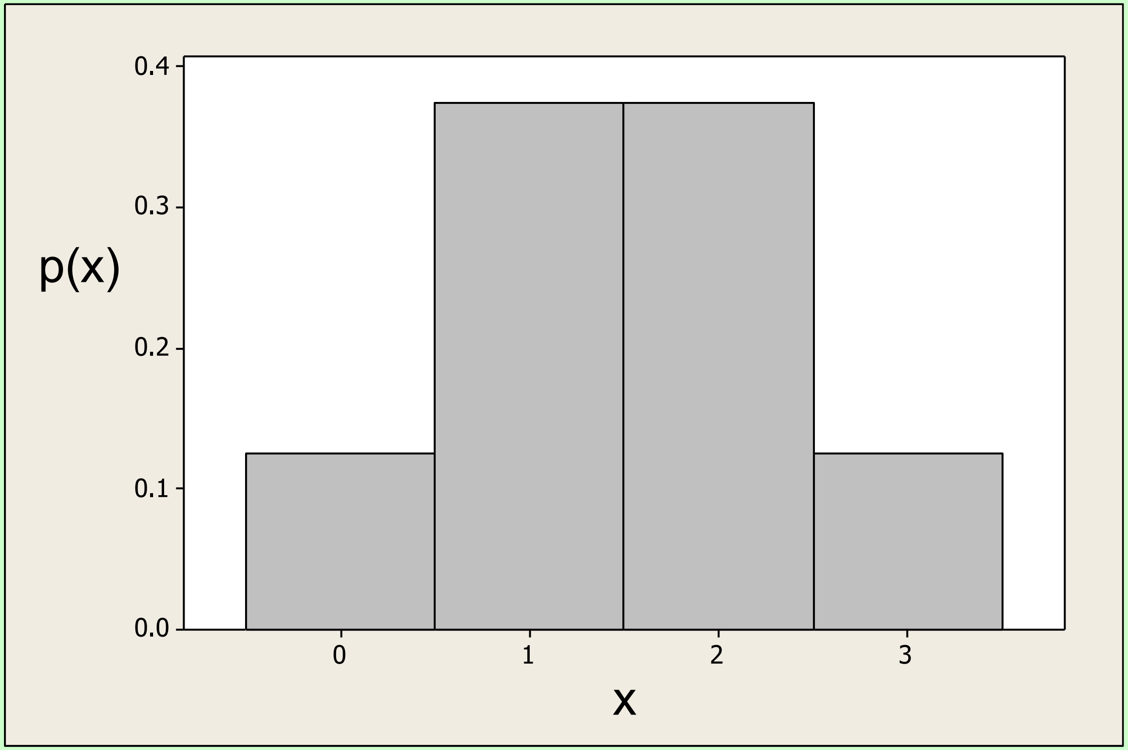
- When X is a discrete RV, we define

$$p(x) = P(X = x)$$

- p(x) is the *probability distribution function* (PDF) of X
  - In the three-tosses example: $p(2) = 3/8$
- Properties of the probability distribution function
  - $0 \le p(x) \le 1$
  - $\sum_{\text{all } x} p(x) = 1$
  - $P(X = a \text{ or } X = b) = p(a) + p(b)$

The Probability Distribution Function

- Graphically:



- Total shaded area is 1

The Cumulative Distribution Function

- The *cumulative distribution function* (CDF) of X is

$$\text{CDF}(x) = P(X \le x)$$

- In the three-tosses example,

$$\begin{aligned} \text{CDF}(2) &= P(X \le 2) \\ &= P(X = 0 \text{ or } X = 1 \text{ or } X = 2) \\ &= p(0) + p(1) + p(2) \\ &= 1/8 + 3/8 + 3/8 \\ &= 7/8 \end{aligned}$$

The Cumulative Distribution Function

- The CDF and PDF are “equivalent” – in the sense that we can construct one from the other
- Example: suppose that X takes the values 10, 20, and 50, and that
  - CDF(10) = 0.3
  - CDF(20) = 0.9
  - CDF(50) = 1

  What is the PDF of X?

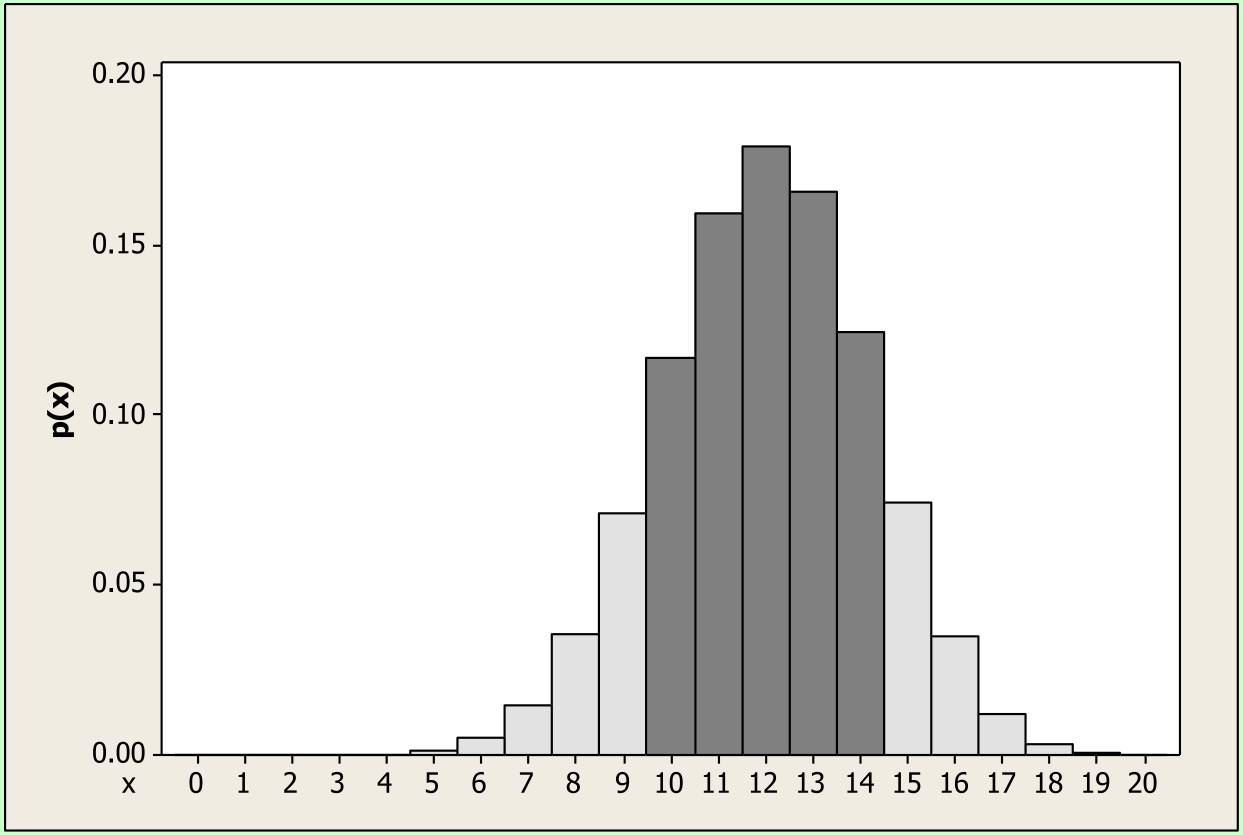
The Cumulative Distribution Function

- Can also find probabilities of "greater than":
  - The complementary event of "$X \le x$" is "$X > x$"
  - Therefore, $P(X > x) = 1 – CDF(x)$
- In the three-tosses example:

$$
\begin{aligned}
P(X > 2) &= 1 – CDF(2) \\
&= 1 – 7/8 \\
&= 1/8
\end{aligned}
$$

- What about $P(a \le X \le b)$?
  - $P(a \le X \le b) = P(X \le b) - P(X < a)$. Do you understand why this is true?

The Cumulative Distribution Function



- $P(10 \le X \le 14) = CDF(14) - CDF(9)$

Expected Value

- How many heads "on average" will we get when tossing a coin three times?
- If we repeat this many times, we expect that
  - In 1/8 of the times, we'll get 0
  - In 3/8 of the times, we'll get 1
  - In 3/8 of the times, we'll get 2
  - In 1/8 of the times, we'll get 3
- So "on average" we'll get

$$0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}$$

Expected Value

- Mathematically: The *expected value* (or *mean*) of a RV X is

$$\mu = E(X) = \sum_{\text{all } x} x p(x)$$

- Sometime write μ_X
- Interpretations of expected value
  - Long run average
  - Fair value of a gamble
- Additivity: $E(X + Y) = E(X) + E(Y)$

Variance and Standard Deviation

- A measure of the variability of a RV is its V*ariance*
- To compute the variance of a discrete RV X
  - Compute μ
  - For each possible x, compute $(x - \mu)^2 \, p(x)$
  - Add up these values
- In a formula: $\sigma_X^2 = \mathrm{Var}(X) = \sum_{\text{all } x} (x - \mu)^2 p(x)$
- Another formula: $\sigma_x^2 = \sum_{\text{all } x} x^2 p(x) - \mu^2$

Variance and Standard Deviation

- In the three-tosses example:

x	0	1	2	3
p(x)	1/8	3/8	3/8	1/8

  - We saw $\mu = 3/2$
  - Using the first formula,

$$\mathrm{Var}(X) = (0 - 3/2)^2 \cdot (1/8) + (1 - 3/2)^2 \cdot (3/8) + (2 - 3/2)^2 \cdot (3/8) + (3 - 3/2)^2 \cdot (1/8) = 3/4$$

  - Using the second formula,

$$\mathrm{Var}(X) = 0^2 \cdot (1/8) + 1^2 \cdot (3/8) + 2^2 \cdot (3/8) + 3^2 \cdot (1/8) - (3/2)^2 = 3/4$$

Variance and Standard Deviation

- Standard deviation (SD) is the square root of the variance:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}$$

- In the three-tosses example: $\sigma = \sqrt{3/4} = 0.866$
- The empirical rule for RVs
  - P(X is between $\mu - \sigma$ and $\mu + \sigma$) is about 2/3
  - P(X is between $\mu - 2\sigma$ and $\mu + 2\sigma$) is about 0.95

Variance and Standard Deviation

- When X and Y are *independent*,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

- Note: it is the variances that add up, not the SDs:

$$\sigma_{X+Y} \neq \sigma_X + \sigma_Y$$

- Both variances and SDs are *always* positive

Covariance

- When X and Y are not *independent*,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$$

$$\begin{aligned}\mathrm{Cov}(X, Y) &= E\left[\left(X - \mu_x\right)\left(Y - \mu_Y\right)\right] \\ &= E\left[XY\right] - \mu_x \mu_Y\end{aligned}$$

$$\rho_{X,Y} = \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

The Binomial Distribution

- Let X = no. of 1's when rolling a die 5 times
- P(X = 2) = ?
- Relevant events (• means "not a 1"):

1 1 • • •	1 • 1 • •	1 • • 1 •	1 • • • 1	• 1 1 • •
• 1 • 1 •	• 1 • • 1	• • 1 1•	• • 1 • 1	• • • 1 1

- No. of these events: $\binom{5}{2} = 10$
- Probability of each such event: $(1/6)^2(5/6)^3$
- Therefore, P(X = 2) = $\binom{5}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3 = 0.161$

The Binomial Distribution

- More generally: $P(X = x) = \binom{5}{x}\left(\frac{1}{6}\right)^{x}\left(\frac{5}{6}\right)^{5-x}$
- Even more generally: When X is the number of "successes" in n independent "trials," each with success probability π,

$$p(x) = P(X = x) = \binom{n}{x}\pi^{x}(1-\pi)^{n-x}$$

- Such an X is called a *Binomial RV* with parameters n and π

The Binomial Distribution

- Binomial examples
  - Number of students in this class who own an iPod
  - Number of no-shows for a flight
    - (Independence?)
  - Number of times next week I'll get stuck in the subway on my way to work

- An important part of understanding probability/statistics is recognizing a "binomial situation"

The Binomial Distribution

- When X is a Binomial RV with parameters n and π,

$$E(X) = n\pi$$

$$Var(X) = n\,\pi(1-\pi), \qquad \sigma_x = \sqrt{n\pi(1-\pi)}$$

- In the "rolling a die 5 times" example (Expected number of successes – number of ones)

$$E(X) = 5\cdot(1/6) = 5/6$$

$$\sigma_x = \sqrt{5(1/6)(5/6)} = 0.833$$

The Binomial Distribution – Example

- A salesperson is about to visit 25 potential customers; the probability of a successful visit (a "sale") is 0.3, independent of other visits. Let X be the number of sales.
  1. $p(8) = P(X = 8) = ?$
  2. $CDF(8) = P(X \leq 8) = ?$
  3. $E(X) = \mu_x = ?$
  4. $SD(X) = \sigma_x = ?$

Sampling with replacement vs Sampling without replacement

- Consider a jar with 20 marbles – 12 red and 8 green.
- You randomly select 5 marbles (one at a time with replacement)
- P(Exactly 3 Red) = $\binom{5}{3} .6^3 \, .4^2$
- What happens if the sampling is done without replacement?

The Hypergeometric Distribution

- N: Population size
- n: Sample size
- r : Number of "successes" in population
- x: Number of "successes" in sample
- $P(X = x) = \dfrac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}$
- Previous Example (w/o replacement)

$$P(X = 3) = \frac{\binom{12}{3}\binom{8}{2}}{\binom{20}{5}}$$

The Poisson Distribution

- Suppose we count events that occur randomly in time, and that
  - Events occur one at a time
  - The numbers of events in non-overlapping periods are independent
- Example: calls arriving to a call center
- Then we may model the number of events in a fixed time interval by a *Poisson RV*

The Poisson Distribution

- The Poisson PDF:

$$p(x) = P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \quad x = 0, 1, 2, \ldots$$

- There is no upper limit on x (unlike binomial)
- λ is the (only) parameter of the distribution. It is also both the mean and the variance of X:

$$E(X) = Var(X) = \lambda$$

- To get λ, often need to multiply a “rate” per time unit by the appropriate duration

The Poisson Distribution – Example

- Phone calls arrive randomly to a call center at a rate of 4 calls per hour. Under the Poisson assumption, find
  1. P(5 calls between 8 and 10) = ?
  2. P(0 calls between 8 and 8:15) = ?
  3. P(At least 1 call between 9 and 10) = ?